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Supplementary Material S1: Expert Usability Assessment Questionnaire

UML-Based Conceptual Modeling Framework Evaluation

Study Title: A Reproducible UML-Based Framework for Conceptual Modeling of *Aedes* Population Dynamics

Part A: Participant Information

1. **Professional Background** (select one):
 - ☐ Ecological modeler
 - ☐ Software engineer with ecological modeling experience
 - ☐ Public health / Vector control specialist
 - ☐ Other: _____
 2. **Years of experience in ecological/population modeling:** _____
 3. **Prior experience with UML** (select one):
 - ☐ None
 - ☐ Basic (seen diagrams, limited use)
 - ☐ Intermediate (used for documentation)
 - ☐ Advanced (regular use in software development)
 4. **Familiarity with mosquito population models** (1-5 scale):

1 = No familiarity ... 5 = Expert

Rating: _____
-

Part B: Baseline Task Assessment

Completed after reviewing the narrative model description (before UML tutorial)

5. **How many distinct model components did you identify?** _____
6. **How many relationships between components did you identify?** _____
7. **How many assumptions (explicit or implicit) did you identify?** _____

8. How confident are you in your understanding of the model structure?

1 = Not at all confident ... 5 = Very confident

Rating: _____

9. What aspects of the narrative description were unclear?

Part C: UML Task Assessment

Completed after the UML tutorial and reconstruction task

10. How many distinct model components did you identify using UML? _____

11. How many relationships between components did you identify using UML? _____

12. How many assumptions (explicit or implicit) did you identify using UML? _____

13. How confident are you in your understanding after using UML?

1 = Not at all confident ... 5 = Very confident

Rating: _____

Part D: Framework Usability Assessment

Rate your agreement with each statement (1 = Strongly Disagree, 5 = Strongly Agree)

#	Statement	Rating (1-5)
14	UML diagrams clarified model structure compared to narrative description	
15	The framework reduced ambiguity in understanding model assumptions	
16	I could use this framework independently after the tutorial	
17	The learning investment is acceptable given the benefits	
18	I would recommend this approach to colleagues	
19	The simplified overview diagram was helpful for initial understanding	
20	The class diagram effectively communicated system structure	
21	The state machine diagram clearly represented the life cycle	
22	The activity diagram helped understand process ordering	

#	Statement	Rating (1-5)
23	The sequence diagram clarified component interactions	

Part E: Comparative Assessment

24. Which diagram type was MOST useful for understanding the model?

- ☐ Simplified overview
- ☐ Class diagram
- ☐ State machine diagram
- ☐ Activity diagram
- ☐ Sequence diagram

25. Which diagram type was LEAST useful or hardest to interpret?

- ☐ Simplified overview
- ☐ Class diagram
- ☐ State machine diagram
- ☐ Activity diagram
- ☐ Sequence diagram

26. Compared to narrative descriptions you typically encounter, the UML framework is:

- ☐ Much less clear
- ☐ Somewhat less clear
- ☐ About the same
- ☐ Somewhat more clear
- ☐ Much more clear

27. The time required to learn the UML framework is:

- ☐ Unacceptable for the benefits provided
- ☐ High but acceptable
- ☐ Moderate and acceptable
- ☐ Low
- ☐ Very low

Part F: Open-Ended Feedback

28. What aspects of the UML framework were most useful?

29. What aspects were most confusing or difficult?

30. What improvements would you suggest for the framework?

31. **Would you use this framework in your own work? Why or why not?**

32. **Any additional comments:**

Part G: Task Completion Record

To be completed by evaluator

Metric	Baseline (Narrative)	UML Task
Components correctly identified	/12	/12
Relationships correctly identified	/8	/8
Assumptions correctly identified	/8	/8
Time to complete task (minutes)		
Errors in interpretation		

Thank you for participating in this evaluation!

Your feedback will help improve the framework for the ecological modeling community.

Document version: 1.0 Date: 2024